

Input&Output Overview

机器人训练后输入虚拟



Leader Arm

Controlled by human.

Record joint poses, called **Action**

Later the model needs to predict this.



Follower Arm

Controlled by program.

Record joint poses and camera images, called **Observation**

Used as input to the model.

In data recording
Follow recorded **Action**, using PID
In testing
Follow predicted **Action**, using PID

Why Action Chunking?

1. Compounding Errors

累积误差问题

small mistakes made by a model in early steps that accumulate over time, leading to significant errors in long-term predictions or decisions.

有效的决策步长

每个时刻观察状态

“reduce the effective horizon of long trajectories”

Single-Step Policy

$$\pi_{\theta}(a_t | s_t)$$

预测 → 再观察
One timestep, one decision made.

Action Chunking Policy

$$\pi_{\theta}(a_{t:t+k} | s_t)$$

k timesteps, one decision made.

观察一次, 产出下k个时刻决策

Lower chance of making mistake.
(Mistake is not fatal, it can be fixed by later action, but continuous high-frequency mistakes are fatal.)

配错不可怕, 高频配错才可怕

Why Action Chunking?

非马尔可夫表现: 当前的动作不仅依赖于当前的状态, 还依赖于过去状态/动作

2. Non-Markov Performance

Non-Markov Behavior

Every task could be designed to follow Markov property, as long as we can extend the state space. However, state space can't contain everything.

e.g.

Pausing while cooking to wait for the oil to heat up.

Markov property

Future state/action only depends on current state.

$$P(X_t | X_{t-1}, X_{t-2}, \dots, X_1) = P(X_t | X_{t-1})$$

$$P(a_t | s_t, s_{t-1}, \dots, s_1) = P(a_t | s_t)$$

Action Chunking Policy could model temporal reliance within chunk.

“Put oil, wait 3 seconds, then ...”

模拟策略内部时序依赖

$$\pi_{\theta}(a_{t:t+k} | s_t)$$

Why Action Chunking?

Single-Step Policy

$$\pi_{\theta}(a_t | s_t)$$

Severe compounding error problem.

Can not model temporal dependence.

Action Chunking Policy

$$\pi_{\theta}(a_{t:t+k} | s_t)$$

Reduce the effective horizon, ease compounding error.

Extend output window to **model temporal dependence.**

History-Conditioned Policy

$$\text{e.g. } \pi_{\theta}(a_t | s_t, s_{t-1}, s_{t-2}, \dots)$$

Include history to handle temporal dependence

Will have Casual Misidentification issue.

找正确原因难度↑
因为找原因范围↑

错误的归因问题

Why Action Chunking?

t	Joint Pose		Obstacle	Action Target	
	Brake	Brake Light		→	Brake
t=0	0	0	0	→	0
t=1	0	0	0.3	→	1
t=2	1	1	0.3	→	1
t=3	1	1	0.4	→	1
...	...	...	...	→	...
t=15	1	1	0.5	→	1
t=16	1	1	0.2	→	1
t=17	1	1	0	→	0
t=18	0	0	0	→	0
t=19	0	0	0	→	0

Casual Misidentification

Wrongly attribute the reason for taking an action to an inappropriate cause. And longer context makes the problem more severe.

Predict by brake light: Accuracy = 90%

Predict by obstacle: Accuracy = 100%



Figure 1: Causal misidentification: more information yields worse imitation learning performance. Model A relies on the braking indicator to decide whether to brake. Model B instead correctly attends to the pedestrian.

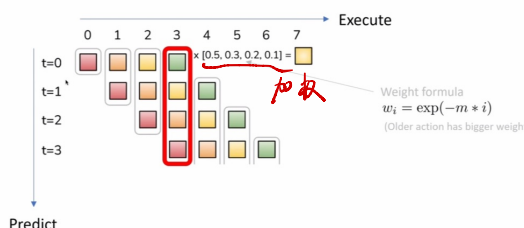
Temporal Ensembling 时序集成

Temporal Ensemble

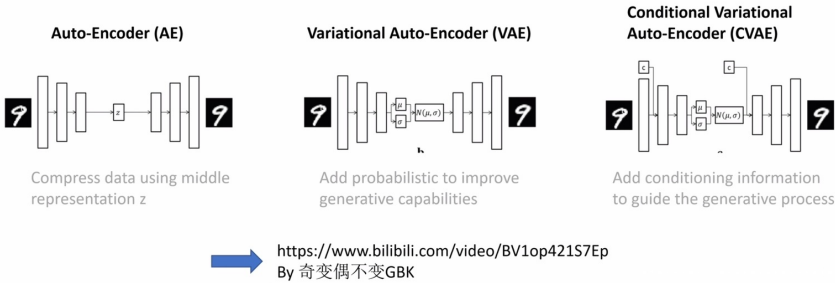
To smooth the output action during inference and in real application.



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Conditional Variational Auto-Encoder (CVAE)



Loss Function

Algorithm 1 ACT Training

- 1: Given: Demo dataset $\mathcal{D}$, chunk size k , weight β .
- 2: Let a_t, o_t represent action and observation at timestep t , $\bar{o}_t$ represent o_t without image observations.
- 3: Initialize encoder $q_\phi(z|a_{t:t+k}, \bar{o}_t)$
- 4: Initialize decoder $\pi_\theta(\hat{a}_{t:t+k}|o_t, z)$
- 5: **for** iteration $n = 1, 2, \dots$ **do**
- 6: Sample $o_t, a_{t:t+k}$ from $\mathcal{D}$
- 7: Sample z from $q_\phi(z|a_{t:t+k}, \bar{o}_t)$
- 8: Predict $\hat{a}_{t:t+k}$ from $\pi_\theta(\hat{a}_{t:t+k}|o_t, z)$
- 9: $\mathcal{L}_{reconst} = \text{MSE}(\hat{a}_{t:t+k}, a_{t:t+k})$
- 10: $\mathcal{L}_{reg} = D_{KL}(q_\phi(z|a_{t:t+k}, \bar{o}_t) \parallel \mathcal{N}(0, \mathbf{I}))$
- 11: Update θ, ϕ with ADAM and $\mathcal{L} = \mathcal{L}_{reconst} + \beta \mathcal{L}_{reg}$

$\mathcal{L}_{reconst}$

Align predicted actions with ground-truth actions.

Principle $\min_{\theta} - \sum_{s_t, a_{t:t+k} \in D} \log \pi_{\theta}(a_{t:t+k} | s_t)$

Implement MSE, L1 Loss, etc.

$\mathcal{L}_{reg}$

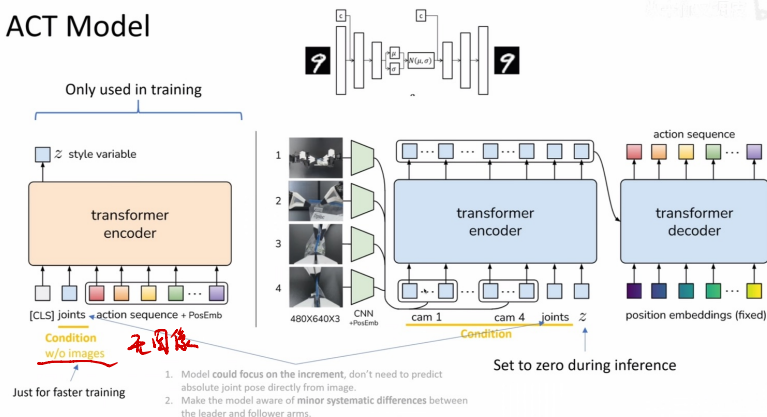
Align the z distribution to Gaussian.

$q(z|x) \sim \mathcal{N}(\mu, \text{diag}(\sigma^2))$

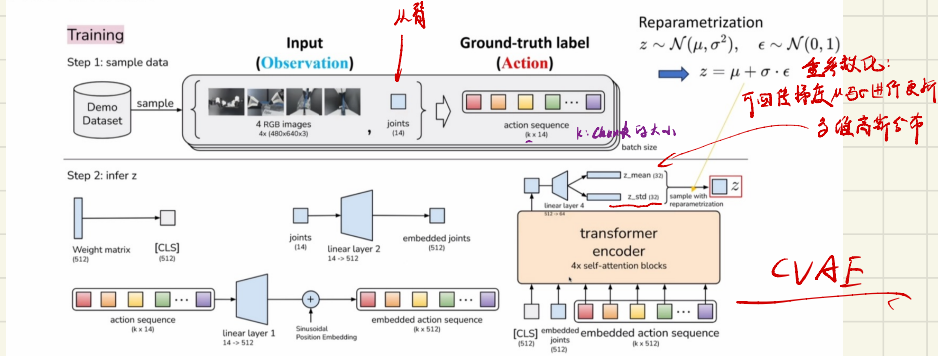
$D_{KL}(q(z|x) \parallel p(z)) = \frac{1}{2} \sum_{i=1}^d (1 + \log(\sigma_i^2) - \mu_i^2 - \sigma_i^2)$

$\mathcal{N}(0, \mathbf{I})$

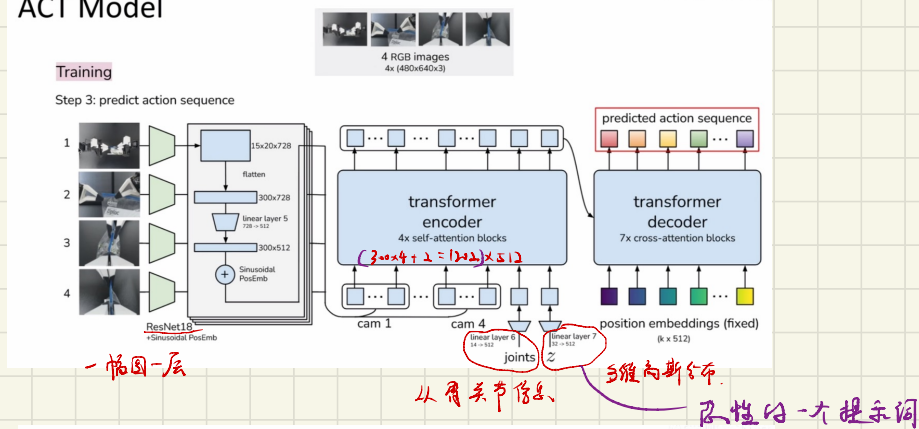
ACT Model



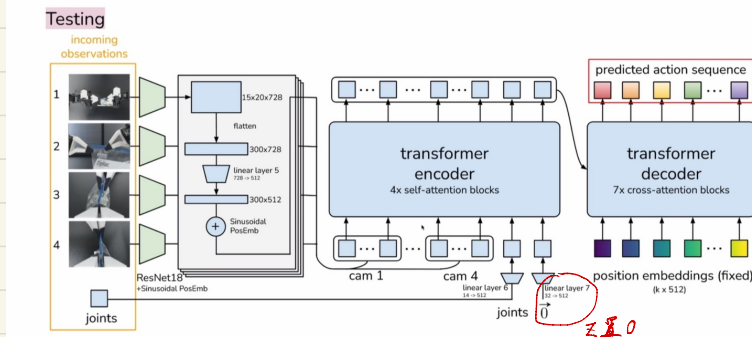
ACT Model



ACT Model



ACT Model



一个任务 → 一个模型 无泛化性.

